https://github.com/chestersanchez36-boop/APM1100-FA8.git

Chester Sanchez 2024050292 FA8

1a) $P(X=0)$ Red: $\binom{15}{4} \div \binom{25}{4} = \frac{1365}{12650}$ Black: $\binom{7}{4} \div \binom{25}{4} = \frac{238}{12650}$

Green: $\binom{18}{4} \div \binom{25}{4} = \frac{3060}{12650}$ $\boxed{P(X=0) = \frac{1}{3}\left(\frac{1365 + 238 + 3060}{12650}\right) = \frac{171}{1265} \approx 0.1352}$

1b) $P(X_i=1)$, $i=1,2,3,4$ Red $= \frac{10}{25}$ Black $= \frac{8}{25}$ Green $= \frac{7}{25}$

$\boxed{P(X_i=1) = \frac{1}{3}\left(\frac{10+8+7}{25}\right) = \frac{1}{3}\left(\frac{25}{25}\right) = \frac{1}{3} = 0.3333}$

1c) $X = X_1 + X_2 + X_3 + X_4$, $E[X_i] = 1 \cdot P(X_i=1) + 0 \cdot P(X_i=0) = P(X_i=1) = \frac{1}{3}$

$E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4] = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{4}{3} \approx \boxed{1.3333}$

2a) $\sum_{n=1}^{\infty} P(X=n) = 1$ $\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$

$\hookrightarrow \sum_{n=1}^{\infty} \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right] = 2\sum_{n=1}^{\infty}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$

$S_N = 2\left[\left(\frac{1}{2} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{12}\right) + \dots + \left[\frac{1}{N(N+1)} - \frac{1}{(N+1)(N+2)}\right]\right]$, $S_N = 2\left[\frac{1}{2} - \frac{1}{(N+1)(N+2)}\right]$

$\sum_{n=1}^{\infty} P(X=n) = \lim_{N\to\infty} S_N = 2\left(\frac{1}{2} - 0\right) = 2 \cdot \frac{1}{2} = \boxed{1}$

2b) $E[X] = 2$ $E[X] \sum_{n=1}^{\infty} n \cdot P(X=n) = \sum_{n=1}^{\infty} n \cdot \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)} = \frac{1}{n+1} - \frac{1}{n+2}$

$E[X] = 4\lim_{N\to\infty}\left[\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left[\frac{1}{N+1} - \frac{1}{N+2}\right]\right] = \left(\frac{1}{2} - \frac{1}{N+2}\right) = 4\left(\frac{1}{2} - 0\right) = 4 \cdot \frac{1}{2} = \boxed{2}$

2c) $E[X] = \infty = \sum_{n=1}^{\infty} n^2 \cdot P(X=n) = \sum_{n=1}^{\infty} \frac{4n^2}{n(n+1)(n+2)}$, $\lim_{n\to\infty}\left(\frac{n}{(n+1)(n+2)} \div \frac{1}{n}\right) = \lim_{n\to\infty} \frac{n^2}{n^2+3n+2} = 1$

$\hookrightarrow 0$ $\rightarrow$ $E[X^2] 4\sum_{n=1}^{\infty} \frac{n}{(n+1)(n+2)} = \boxed{\infty}$

3) $E[X] = \frac{2}{5}$, a & b? $\int_0^1 (a + bx^2)\,dx = 1$, $\left[ax + \frac{b}{3}x^3\right]_0^1 = 1$, $a + \frac{b}{3} = 1$

$\int_0^1 x(a + bx^2)\,dx = \frac{2}{5} = \int_0^1 (ax + bx^3)\,dx = \frac{2}{5}$, $\left[\frac{a}{2}x^2 + \frac{b}{4}x^4\right]_0^1 = \frac{2}{5}$, $\frac{a}{2} + \frac{b}{4} = \frac{2}{5}$

$a = 1 - \frac{b}{3}$, $\frac{1}{2}\left(1 - \frac{b}{3}\right) + \frac{b}{4} = \frac{2}{5}$, $\frac{1}{2} - \frac{b}{6} + \frac{b}{4} = \frac{2}{5}$, $\frac{6}{12} + \frac{-2b+3b}{12} = \frac{2}{5}$, $\frac{b}{12} = \frac{2}{5} - \frac{1}{2} = \frac{1}{10}$

$b = \frac{12}{10} = \boxed{\frac{6}{5}}$, $a = 1 - \frac{1}{3}\left(\frac{6}{5}\right) = 1 - \frac{2}{5} = \boxed{\frac{3}{5}}$

4) $Y = \frac{X-\mu}{\sigma}$ $E[Y] = E\left[\frac{X-\mu}{\sigma}\right] = \frac{1}{\sigma}E[X-\mu] = \frac{1}{\sigma}(E[X] - \mu) = \frac{1}{\sigma}(\mu - \mu) = 0$

$\mathrm{Var}(Y) = \mathrm{Var}\left(\frac{X-\mu}{\sigma}\right) = \mathrm{Var}\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \left(\frac{1}{\sigma}\right)^2 \mathrm{Var}(X) = \frac{1}{\sigma^2} \cdot \sigma^2 = 1$

5a) X - Poisson$(\lambda_1 = 2)$: $M_X(t) = e^{2(e^t - 1)}$, Y - Poisson$(\lambda_2 = 3)$: $M_Y(t) = e^{3(e^t - 1)}$

$M_W(t) = M_X(t) \cdot M_Y(t) = e^{2(e^t-1)} \cdot e^{3(e^t-1)} = \boxed{e^{5(e^t-1)}}$

5b) $M_W(t) = e^{5(e^t-1)} = e^{\lambda(e^t-1)}$, $\boxed{W = \text{Poisson distribution \& } \lambda = 5}$

5c) $M'_W(t) = \frac{d}{dt}\left[e^{5(e^t-1)}\right] = 5e^t \cdot e^{5(e^t-1)}$, $M'_W(0) = 5e^0 \cdot e^{5(e^0-1)} = 5 \cdot e^{5 \cdot 0} = 5$

$M''_W(t) = \frac{d}{dt}\left[5e^t \cdot e^{5(e^t-1)}\right] = (5e^t)(e^{5(e^t-1)}) + (5e^t)(5e^t \cdot e^{5(e^t-1)})$

$M''_W(t) = 5e^t e^{5(e^t-1)}(1 + 5e^t)$

$M''_W(0) = 5e^0 e^{5(e^0-1)}(1 + 5e^0) = 5 \cdot 1 \cdot e^{5 \cdot 0} \cdot (1+5) = 5 \cdot 1 \cdot 1 \cdot 6 = \boxed{30}$